Agenda





- * Basic Definitions
  - Sample space
  - Event
- * Set operations
  - Union, Intersection, Complement
- * Probability
- * Mutually Exclusive Events

=> When was the last time ?

17 year

=> Probability in the context of D.S ?

=>

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Sample space: Collection of all possible outcomes of an experiment

Ex: Throwing a Dice

$S = \{1, 2, 3, 4, 5, 6\}$ ⇒ Unique

Event: Any subset of the sample ~~set~~ space

(or)

Any collection of outcomes.

Ex: odd number = $\{1, 3, 5\}$

Even number = $\{2, 4, 6\}$

prime no = $\{2, 3, 5\}$

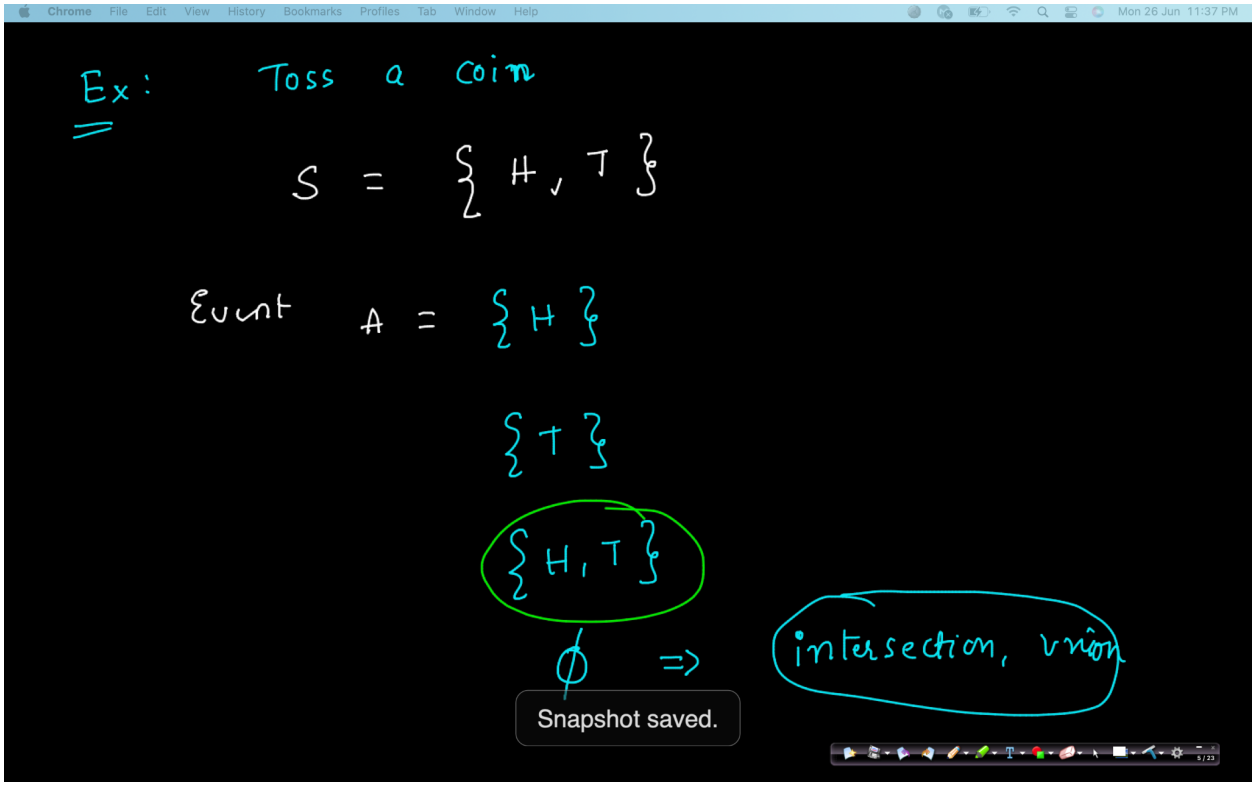
$A = \{1, 2\}$

Greater than 3 = $\{4, 5, 6\}$

$C = \{1, 3, 5, 7\}$

$\{1, 2, 3, 4, 5, 6\}$

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Mon 26 Jun 11:37 PM
Ex: Toss a coin
S = {H, T}
Event A = {H}
{T}
{H, T}
∅ ⇒
intersection, union
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✓ Two coin tosses ?

$S = \{(H, H), (H, T), (T, H), (T, T)\}$

Event:

atleast one head $= \{HH, HT, TH\}$

Both tosses are same $= \{HH, TT\}$

Sample space, Event

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Quiz #1

$S = \{1, 2, 3, 4, 5, 6\}$

a. $\{1\}$ ✓

b. $\{1, 3\}$ ✓

c. $\{1, 3, 5\}$ ✓

d. $\{1, 3, 5, 7\}$

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Quiz #2 ✓ tossing coin followed by a dice.

How many elements will be there in sample space?

1, 2, 3, 4, 5, 6

$S = \{(H,1), (H,2), (H,3), (H,4), (H,5), (H,6)$

$(T,1), (T,2) \quad (T,3), (T,4), (T,5) (T,6)\}$

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✓ Dhoni and Sixes (super over)

1 ball — possible outcomes — hit a six = {S, F}

2 balls — {SS, SF, FS, FF}

3 balls — {SSS, SSF, SFS, FSS, ---- }

Six not

4 balls ⇒ 2^4 = 2 x 2 x 2 x 2 = 16

2^3 ⇒ 8

n balls ⇒ 2^n

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Set operations : Intersection, Union, Complement

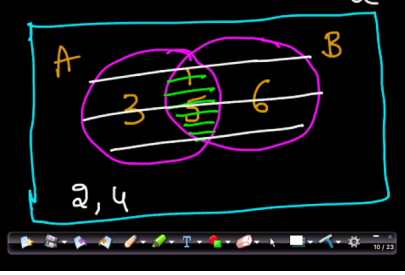
Dice : $S = \{1, 2, 3, 4, 5, 6\}$

$A = \{1, 3, 5\}$ ✓

$B = \{1, 5, 6\}$ ✓

Q When will both A & B will win ?

$A \cap B = \{1, 5\}$



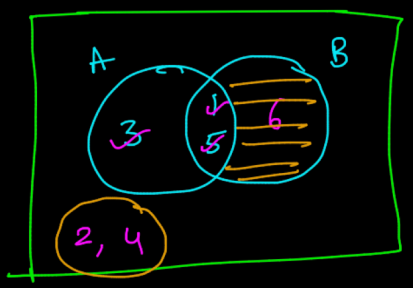
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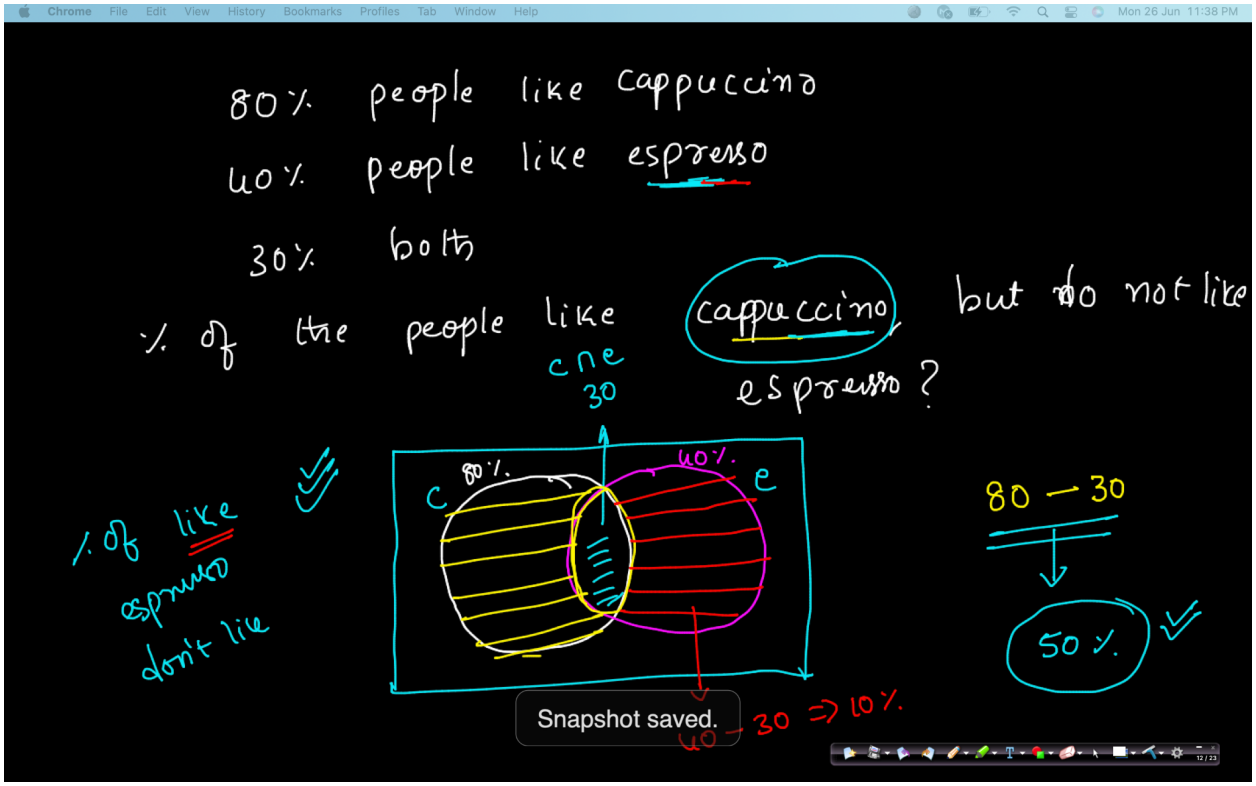
A (or) B will win ?

$A \cup B = \{1, 3, 5, 6\}$

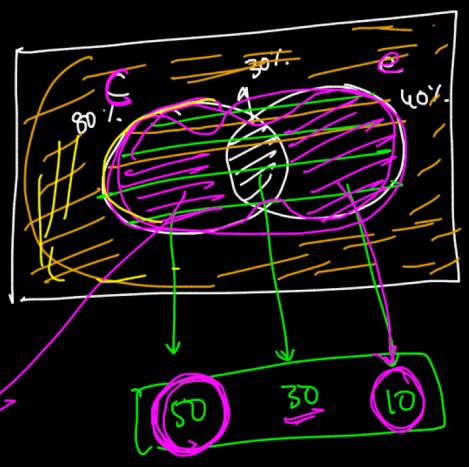
⇒ When will A loose ?

$A^c = \{2, 4, 6\}$




80% people like cappuccino
40% people like espresso
30% both
% of the people like cappuccino but do not like espresso ?
one 30
% of like espresso don't like
80%
40%
C
E
80 — 30
50 %
40 — 30 ⇒ 10%
Snapshot saved.

neither C nor e



a. 0

b. 10

c. 40

80% C

40% e

30% Both

100%

100 − 90 ⇒ 10%

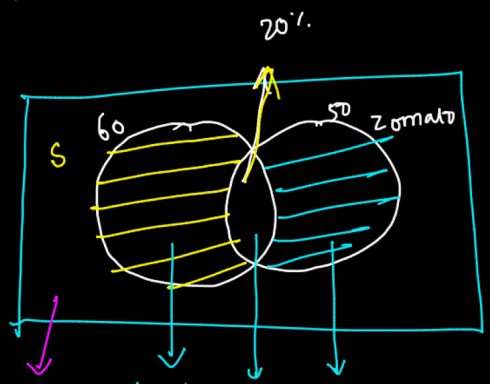
Tea

60% use Swiggy

50% zomato

20% both

% use swiggy, but do not use zomato?



60% − 20%

40%

= 90%

100 − 90%

= 10%

Coin : ⇒ what is the probability getting H

{H, T}

P(H) = 1/2 = 0.5

P(T) = 1/2 = 0.5

0 <= P <= 1

0% 100%

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Dice :

{1, 2, 3, 4, 5, 6}

P({1}) = 1/6

P({6}) = 1/6

Snapshot saved.

Dice $S = \{1, 2, 3, 4, 5, 6\}$

$A = \{2, 4, 6\}$

$B = \{1, 2\}$

① $P(A) = 3/6 = 1/2$

② $P(B) = 2/6 = 1/3$

③ $P(A^c) = 3/6 = 1/2$

$A^c = \{1, 3, 5\}$

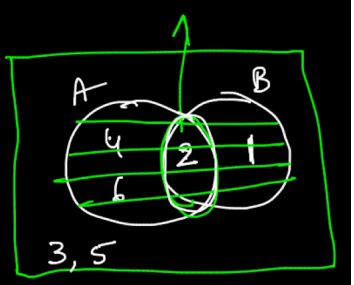
④ $P(B^c) = 4/6 = 2/3$

$B^c = \{3, 4, 5, 6\}$

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⑤ $P(A \cap B) = 1/6$

$A \cap B = \{2\}$



⑥ $P(A \cup B) = P(A) + P(B)$

$= \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

1 $\{1, 2, 4, 6\}$ $\Rightarrow$ $4/6 = 2/3$ ✓

$\{2, 4, 6\}$ $\{1, 2\}$ $\{2\}$

** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Mutually Exclusive

$C = \{1, 3, 5\}$

$A = \{2, 4, 6\}$

$B = \{1, 2\}$

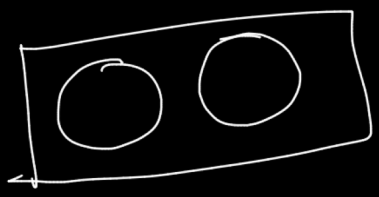
$P(A \cap C) = 0$

$\Downarrow$

A & C are mutually Exclusive

(&)

Disjoint sets



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Bin

ME sets?

a. Youtube premium vs Non premium

cappuccino vs Espresso

Swiggy vs Zomato

Amazon vs Flipkart

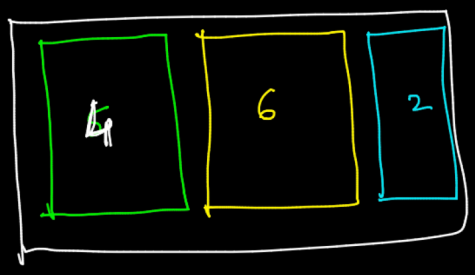
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4 green balls

6 yellow balls

2 Blue balls

$P(Y \cup B)$



a. 4/12

b. 6/12

c. 8/12 ✓

d. 10/12

$P(Y) = 6/12$ $P(B) = 2/12$

$P(G) = 4/12$ $P(Y \cap B) = 0$

$P(A \cup B) = P(Y) + P(B) - P(Y \cap B) = 8/12$

$= 6/12 + \frac{2}{12} - 0$

$$P(A \cup B) = P(A) + P(B)$$

⇒ In an NPS survey, It is seen that 90 % are either promoters or neutral. 30 % are neutral (or) detractors. what percent of people are neutral ?

Snapshot saved.



